

1448/03/20 هـ	تاريخ التطبيق	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج
2027/09/02 م	Date Effective			Form Title
2.0	رقم النسخة			رمز النموذج
إدارة التقنية	مالك النموذج		Apexium	جهة الإصدار
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Requester information		
Requester Name:	General Department:	
Email:	Section:	
Phone Number/extension:	Force or BU:	
Strategic Information (Filled By Project Owner)		
Initiative Name	Priority	
Project Name, from Roadmap	SIMF 4th Edition 2026	
Components from To-Be Architecture Realized		
Screening Questions (Filled by Technology)		
Requirements/SRS/User Stories Defined and Approved	Defined. The Business Requirements Document delivered with this design carries 132 numbered, individually testable requirements. The Solution Design Document, also delivered with this design, traces every one of those identifiers to the module that implements it and to its verification method. Approval is the Ministry's.	
Integration Requirements Defined and Approved	Defined. Section 2.4 gives every interface with its protocol, port, payload and authentication, and section 2.8 carries the Communication Requirements Matrix. One outbound integration remains pending its firewall exception.	
Security Requirements Defined and Approved	Defined. Section 2.9 Solution Security View, aligned to the National Cybersecurity Authority Secure Application Development Standard and to the Personal Data Protection Law. Open items are listed in section 3.	
Uses approved Architecture/Design Methodology	Modular monolith with Domain Driven Design layering and defence in depth, described in sections 2.1 and 2.2, with the alternatives considered and rejected recorded in the section 2.1 decisions table. Confirmation against Technology's approved methodology rests with Technology.	
Architecture Constraints Documented	Yes. The Architecture Constraints row of this section states them, and sections 1.2, 2.1 and 2.7 carry the hosting, zone and network constraints that follow from them.	
Uses Approved Solution Stack (Hardware/Software)	The stack is stated in the Software Specification and Hardware Specification rows of this section, with per node sizing in the section 2.1 table. Confirmation against Technology's approved stack rests with Technology.	
Is system described sufficiently (Section 1.1 to 1.3))	Yes. Section 1.1 gives the purpose, scope and glossary, section 1.2 the goals, quality goals, assumptions and non-goals, and section 1.3 the stakeholders, their concerns and the system boundary.	

1448/03/20 هـ	تاريخ التطبيق	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج
2027/09/02 م	Date Effective			Form Title
2.0	رقم النسخة			رمز النموذج
إدارة التقنية	مالك النموذج		Apexium	Form Code
	Form Owner			جهة الإصدار
				Issuer

Is solution design described sufficiently (Section 2.1 to 2.9)	Yes. Sections 2.1 to 2.9 are complete, with three figures, the per tier sizing table, the architecture decisions table, the interface table and the Communication Requirements Matrix.
Is the Risk and Mitigation Strategies provided	Yes. Section 3 gives the risk register with impact, likelihood, mitigation and owner for each risk, and the open items register with a closing condition for every item still outstanding.
Document Completion Approved?	For Technology. This row records the review board's own verdict, and is the same decision the Approvals block signs.

Document identification	Document SIMF-HLD-005, version 2.0, status Issued for review. Apexium issues and authors it for the Royal Saudi Naval Forces (RSNF) under a Ministry of Defense programme, roadmap project SIMF 4th Edition 2026. It conforms to IEEE 1016-2009 and ISO/IEC/IEEE 42010:2011. Issue date is TBD, owner Apexium, recorded in Hijri and Gregorian at issue.
BRD (Business Requirements Document)	A Business Requirements Document is delivered with this design and records the business objectives, scope, stakeholders and business rules. The Solution Design Document traces every requirement identifier to the module that implements it.
SRS/ User Stories	The functional, non-functional and operational requirements are carried in the Business Requirements Document rather than a separate specification. The Solution Design Document carries the traceability and the verification method for each of them.
Solution Requirements	The platform is single-tenant and open to public and international registration. Four client applications share one backend API and two databases. Sections 1.2 and 1.3 give the goals and the system boundary; section 2.2 names the modules.
Integration Requirements	Section 1.3 fixes the system boundary and names every external and internal system the platform depends on. Section 2.4 defines each interface, with its direction, protocol and port, authentication and error behaviour, and marks the interfaces still to be agreed.
Security Requirements	The platform runs its own identity store , and sign-in uses an e-mail address, a password and a second factor . Permissions are granted per page and per action, and confidential columns and files use AES-256-GCM with keys in a key management service. Section 2.9 carries the full posture and the alignment to the Saudi Personal Data Protection Law (PDPL).
Hosting Requirements	Hosting is on premises on the SITE Private Cloud, on Windows Server 2022 with SQL Server 2022. Two security areas contain four network zones. Figure 1 and section 2.1 give the network layout; section 2.7 maps components onto environments.
Architecture Methodology or Best Practice used	Section 2.1 records the architectural style and the decisions taken with their rejected alternatives; section 2.2 sets out the layering and the module boundaries. The design follows defence in depth, keeps the application tier stateless and makes the database tier highly available.
Architecture Constraints	The forum runs on fixed dates, the platform freezes before it opens, and an outage cannot be rescheduled. There is no public-cloud dependency, and the source code is handed over to the customer. Section 1.2 numbers the constraints; section 2.8 states the two-database separation invariants.
Software Specification	The stack is .NET 10 with FastEndpoints, FluentValidation and EF Core in code-first mode. The front ends use Blazor Server and Blazor SSR over the shared SIMF.Components library, with Flutter and Dart for Android and iOS. IIS hosts the web tier, Serilog carries logging, and section 2.2 assigns each part to its module.
Hardware Specification	The figures below are the hardware agreed with the customer. Production runs four API hosts, two website hosts, two Control Panel hosts, two SIMF.MobileEdge hosts and two database hosts. Traffic passes two active/passive pairs. A web application firewall (WAF) and load balancer pair sits at the internet edge. An API load balancer pair sits in front of the application tier. Section 2.1 gives per-node sizing; the mobile edge and MinIO figures are a supplier proposal in the section 3 register.

1448/03/20 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/09/02 م	Date Effective			رمز النموذج Form Code
2.0	رقم النسخة Rev. Number			جبهة الإصدار Issuer
إدارة التقنية	مالك النموذج Form Owner		Apexium	

1.1 1.1 Purpose

This document is the high-level design of the platform built for the Saudi International Maritime Forum (SIMF). It addresses the Ministry of Defense technology review board and the Royal Saudi Naval Forces (RSNF) as system owner. It also serves the teams that build, host and support the platform. Section 1.3 names each stakeholder and their concerns.

The description conforms to IEEE 1016-2009 for software design descriptions and to ISO/IEC/IEEE 42010:2011 for architecture description.

This document carries no database schema, no interface specification and no unit-level logic. The Solution Design Document, delivered alongside it, carries all three.

The Requester Information, Strategic Information and Approvals blocks are Ministry-owned; the Ministry completes them on receipt.

Glossary

Each acronym below carries its expansion and its role in the design.

Acronym	Expansion	Role in this design
SIMF	Saudi International Maritime Forum	The forum and its platform
RSNF	Royal Saudi Naval Forces	System owner
SITE	A company name, not an abbreviation	Hosts the platform, and supplies the AI model and the e-mail service
SSA	Secure Service Area	Outer security area
HSA	High Security Area	Inner security area
NCA	National Cybersecurity Authority	Cybersecurity regulator
ECC	Essential Cybersecurity Controls	NCA control baseline
PDPL	Personal Data Protection Law	Saudi data protection law
OWASP	Open Worldwide Application Security Project	Application security baseline
WAF	Web application firewall	Perimeter control
KMS	Key management service	Holds encryption keys
HSM	Hardware security module	Backs the KMS
AG	Availability Group	Database high availability
TDS	Tabular Data Stream	SQL Server wire protocol
API	Application programming interface	Backend service surface
REST	Representational State Transfer	Interface style for the API
RTO	Recovery time objective	Availability commitment
RPO	Recovery point objective	Availability commitment
OTP	One-time password	Second factor
TOTP	Time-based one-time password	Second factor
DDD	Domain-Driven Design	Backend layering
TBD	To be determined	Open item with a named owner

1448/03/20 هـ	تاريخ التطبيق	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج
2027/09/02 م	Date Effective			Form Title
2.0	رقم النسخة			رمز النموذج
إدارة التقنية	مالك النموذج		Apexium	جهة الإصدار
	Form Owner			Issuer

1.2 System Overview

The Saudi International Maritime Forum (SIMF) platform is a single-tenant event platform. It serves the Royal Saudi Naval Forces (RSNF) under a Ministry of Defense programme, and the current edition is SIMF 4th Edition 2026. *Any member of the public may register*, in the Kingdom or abroad, subject **to administrator approval**. The platform verifies each registrant, issues a digital badge and controls entry by gate scan. It publishes the programme, takes seat reservations, sends reminders, streams live sessions and moderates audience questions. It presents the exhibition, sponsors, media partners, news and the past-edition archive, and it supports networking and meeting requests. It collects ratings, feedback and statistics, and organisers administer the whole event from the Control Panel without a code release.

Attendance at the forum is free. The platform holds no payment, ticketing, invoicing or finance capability, and none is planned.

Quality goals

Four quality goals shape the design, each stated with the scenario that measures it.

Quality goal	Measurable scenario	Target
Availability across the forum days	Attendees, gate operators and organisers use the platform throughout the three forum days.	The availability objectives in section 2.9
Gate response under peak arrival	An operator scans a badge at the gate while arrivals peak.	250 ms at the 95th percentile for a gate scan, at 3,000 concurrent users
Confidentiality of personal data	A reader reaches database rows or stored files without holding the encryption keys.	Identity numbers, mobile numbers and identity-document images stay unreadable; controls in section 2.9
Feature freeze before launch	The forum opens on its fixed dates.	Feature-complete and frozen before opening day, with defect corrections only after that point

Assumptions and constraints

1. The forum runs on fixed dates, so the delivery date cannot move.
2. An outage inside the event window cannot be rescheduled.
3. Hosting stays on premises, with no public-cloud dependency.
4. The source code passes to the customer at handover, as the contract requires.

1448/03/20 هـ	تاريخ التطبيق	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج
2027/09/02 م	Date Effective			Form Title
2.0	رقم النسخة			رمز النموذج
	Rev. Number			Form Code
إدارة التقنية	مالك النموذج		Apexium	جهة الإصدار
	Form Owner			Issuer

1.3 System Context

At this level the platform is a black box: named users on one side, named systems on the other.

Four client applications sit on that boundary: the Flutter mobile app for Android and iOS, the public website, the Control Panel, and the on-site SIMF.BadgeDesk. The mobile app reaches the platform through SIMF.MobileEdge rather than the backend API.

Nine stakeholder groups hold a distinct interface and a distinct concern.

Stakeholder	Interface	Primary concern
Attendees and VIP delegates	Mobile app	Purpose: registration, badge, seats, live participation
Anonymous public	Public website	Purpose: programme and visit information, no account
Administrators and organisers	Control Panel	Suitability: run the event without a code release
Scientific Committee	Control Panel	Suitability: the published forum programme
Gate operators and session moderators	Mobile app and Control Panel	Feasibility: scan throughput, moderated questions
Badge desk staff	SIMF.BadgeDesk	Feasibility: badges issued at the venue
RSNF, the Royal Saudi Naval Forces	Owner; receives platform and source code	Purpose and lifecycle risk: freeze, handover
SITE, the hosting provider	Private cloud, mail relay, inference server	Feasibility within supplied capacity
Smart Time, the official event manager	The forum and its editions	Maintainability and evolvability across editions

One external system and six internal systems complete the context.

System	Boundary	Direction	Channel
YouTube	External	Outbound only	HTTPS caption fetch; playback on the attendee device
SMTP mail relay	Internal, on site	Outbound from the API	SMTP, queued and asynchronous
MinIO object storage	Internal, on site	From the API only	S3 API
GPT OSS 120B inference endpoint	Internal, on site	From the API only	OpenAI-compatible HTTPS, traffic stays on site
SQL Server 2022	Internal, on site	From the API only	TLS-protected SQL client connection
Ministry log collector	Internal	Outbound from each host	Syslog; the host ships the rolling files
Key management service, hardware security module	Internal	Read at node start-up	Product selection pending, section 2.4

No external system initiates an inbound connection to the platform. Client applications on the internet connect inbound through the WAF and the load balancer. Section 2.8 states the protocol and port for each connection.

1448/03/20 هـ	تاريخ التطبيق	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج
2027/09/02 م	Date Effective			Form Title
2.0	رقم النسخة			رمز النموذج
إدارة التقنية	مالك النموذج		Apexium	جهة الإصدار
	Form Owner			Issuer

2.1 Architecture Overview

Architectural style and tiers

The platform is a modular monolith with domain-driven design (DDD) layering. One backend application, SIMF.Api, serves the mobile application, the public website and the Control Panel. Persistence is split across two databases, SIMF_Identity and SIMF_App. Hosting is on premises on Windows Server 2022 with SQL Server 2022. The deployment is three-tier, and in production each tier occupies its own servers. The presentation tier holds SIMF.Web, SIMF.ControlPanel and SIMF.MobileEdge. The application tier is four stateless SIMF.Api hosts. The data tier is SQL Server with the MinIO object store. No presentation project holds a database connection string, so a compromised presentation host gives no direct route to the data. Development and Test run four sites on one node: the API, the Control Panel, the website and the mobile edge.

Security areas, zones and firewalls

Two security areas, SSA and HSA, hold four network zones. Both full names are TBD, owner the site network team, closing condition sign-off of the deployment network design. SSA holds three zones: access, perimeter and presentation. It is the only security area with internet access. The access zone holds attendee devices and administrator browsers. The perimeter zone holds the combined web application firewall (WAF) and load balancer. The presentation zone holds the presentation tier alone. HSA is a single zone carrying internal traffic only, and it holds the application and data tiers. Nothing in HSA has a route to the internet.

Two firewalls stand at zone boundaries. The perimeter firewall guards the internet-facing boundary of SSA. It admits inbound client traffic and permits one outbound destination, the caption fetch described in section 2.4. Its port-level rule set is TBD, owner the site network team, closing condition sign-off of the deployment network design. The internal firewall separates the presentation zone from HSA and is the only route in. It permits TCP 443, and only from the presentation zone. It is also the only barrier in front of the data tier.

Figure 1 answers where each component runs and which boundary each connection crosses. It draws one box per hosting server and labels every arrow with its protocol and port, which section 2.8 lists in full.

Capacity basis

The forum is time-boxed, so sizing targets the event-day peak, not an average load. The design assumes about 5,000 platform users a day and about 1,000 attendees on site, and the largest hall seats 500. A peak factor of 0.6 gives about 3,000 concurrent users, with headroom to about 4,000. At that load the 95th percentile targets are 400 ms for sign-in, 250 ms for a gate scan and 300 ms for a read. Those targets hold while processor use stays below 75 percent. Live session video reaches the attendee device from YouTube, and section 2.4 records that path, so the tiers are sized for REST traffic alone. Section 2.9 sets out scale-out and overload behaviour.

The following table gives each tier with its zone, node count, specification and purpose.

Tier	Zone	Instances	Specification per node	Purpose
WAF and load balancer	SSA perimeter	2, active/passive	Appliance or virtual machine	Terminates TLS, filters against the Open Web Application Security Project (OWASP) rule set, balances to presentation
SIMF.MobileEdge	SSA presentation	2	8 vCPU / 16 GB / 100 GB, proposed minimum	Presentation tier for the mobile application, which never calls the API directly

1448/03/20 هـ	تاريخ التطبيق	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج
2027/09/02 م	Date Effective			Form Title
2.0	رقم النسخة			رمز النموذج
إدارة التقنية	مالك النموذج		Apexium	Form Code
	Form Owner			جهة الإصدار
				Issuer

SIMF.Web	SSA presentation	2	16 vCPU / 32 GB / 300 GB	Public website, server-side rendered, read-heavy
SIMF.ControlPanel	SSA presentation	2	16 vCPU / 32 GB / 300 GB	Administrative Control Panel; needs session affinity
API load balancer	HSA	2, active/passive	Appliance or virtual machine	Distributes requests across the four API nodes
SIMF.Api	HSA	4	16 vCPU / 32 GB / 300 GB	Stateless REST application tier
Language model server	HSA	TBD, owner SITE	Supplied by SITE	Hosts the GPT OSS 120B model behind an OpenAI-compatible interface
Mail server	HSA	TBD, owner SITE	Supplied by SITE	On-site SMTP relay, STARTTLS on port 587
Database	HSA	2	8 vCPU / 64 GB / 2 TB	Hosts both databases; section 2.9 covers availability
MinIO object store	HSA	2	8 vCPU / 16 GB / 4 TB, proposed minimum	S3 file store, reached only through the API

The following table records the architectural decisions and the alternatives rejected.

Decision	Chosen	Rejected alternative	Reason
Application structure	Modular monolith with DDD layering	Microservices	The API is stateless, so event-day capacity comes from adding nodes, not from splitting services.
Persistence	Two databases, SIMF_Identity and SIMF_App	One shared database	Identity data stays in its own store; section 2.8 states the separation invariants.
AI inference	On-site GPT OSS 120B model	A cloud AI service	The call stays inside HSA and generates no internet traffic, so no attendee data leaves the site.
Live video	YouTube plays to the attendee device	Relaying video through the backend	A relay would need a tier sized for video bandwidth on event day.
Second-factor delivery	One-time password (OTP) by e-mail	SMS	E-mail removes any dependency on an SMS supplier and reaches international attendees without a local number.
Live updates	Bounded client polling	A server push transport	Polling rides the sized REST path and holds no long-lived connection per attendee.
Scheduled work	Jobs run in process on the API tier	A separate worker tier	No fourth tier to size, deploy or fail over.

1448/03/20 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/09/02 م	رقم النسخة Rev. Number			رمز النموذج Form Code
2.0	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer
إدارة التقنية				

Identity	Own identity store serving visitors and administrators	Ministry directory integration or single sign-on federation	The users are members of the general public and international attendees, and hold no ministry account to federate.
Database edition	SQL Server 2022 Enterprise	Standard edition	Standard edition offers no readable secondary; section 2.9 sets out the availability design.

2.2 Modules

Client applications

- Mobile application: Flutter on iOS and Android, serving attendees. It reaches the backend through SIMF.MobileEdge rather than directly.
- Control Panel: Blazor Server with the SIMF.Components library, serving administrators.
- Public website: Blazor SSR, carrying forum and programme information. It has no sign-in, registration or account page and persists no personal data. Two pages differ. The visitor assistant relays question text to the artificial intelligence service without retaining identity. A meeting-confirmation link is private to one speaker.
- Badge desk: SIMF.BadgeDesk, a WinForms application that prints badges on site. It works offline with no database connection, uploading each shift over one batch endpoint on the administration surface.

Backend layers

The backend is one process in four layers. SIMF.Api is the boundary, holding endpoints, middleware, authentication and authorisation policies. SIMF.Application holds use-case orchestration and the service abstractions. SIMF.Infrastructure implements those abstractions, covering persistence, storage, e-mail, identity and the audit interceptors. SIMF.Domain holds entities, aggregates, enumerations and rules, carries no framework dependency and depends on no other layer. Dependencies run inwards, from the boundary towards the domain.

Bounded contexts

The backend divides its work into bounded contexts, grouped by area.

Area	Contexts	Covers
Identity	Authentication and registration; account and profile	Sign-up, credentials, profiles
Programme	Programme; bookings; meetings	Sessions, speakers, presentations, summaries, recordings
Venue	Gates and scanning; exhibition; sponsors	On-site entry, booths
Content	News and media; archive; notifications; networking	Published content, archived material, notifications, connections
Platform	Statistics; system configuration; organisation profile; reference data	Reporting, settings, lookups
Artificial intelligence	AI	Subtitle import, summary drafting, question filtering, assistants

1448/03/20 هـ	تاريخ التطبيق	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج
2027/09/02 م	Date Effective			Form Title
2.0	رقم النسخة			رمز النموذج
إدارة التقنية	مالك النموذج		Apexium	Form Code
	Form Owner			جهة الإصدار
				Issuer

Every AI feature runs against the on-site model server in section 1.3. Section 2.8 describes the databases behind these contexts.

Interfaces and scheduled work

One API process publishes two separately gated surfaces. One serves the attendee clients, the mobile application and the public website. The other serves administration, the Control Panel and the badge desk. Both share one response envelope, one error model and one permission model.

The scheduled jobs run inside the API host; there is no worker server. They sweep dormant accounts, close the registration gate and release unclaimed seats before a session starts. Others send session and meeting reminders, issue rating prompts and dispatch e-mail. One node is elected to run the database-driven jobs, so a reminder is issued once for the tier, not per node. If it stops, another takes over within one polling interval. E-mail dispatch runs on every node, each draining its own queue.

2.3 System Workflow

Seven workflows are architecturally significant. Figure 2, the system and data flow diagram, shows the processes, stores and trust boundaries these workflows cross. Figure 3, the end-to-end sequence, orders the exchange between the clients, the API and the stores. It covers registration and approval, sign-in with the second factor, and the gate scan.

Registration and approval

1. The visitor signs up in the mobile application.
2. The visitor confirms the e-mail address with a one-time password (OTP).
3. The visitor completes the profile.
4. An administrator reviews the registration in the Control Panel and approves or rejects it.
5. Approval issues a digital quick-response (QR) badge and notifies the visitor.

Sign-in with the second factor

1. The user submits an e-mail address and a password.
2. The user supplies the second factor, which section 2.9 defines.
3. The API validates the factor and returns a short-lived access token with a rotating refresh token.
4. A mobile user may then enrol a biometric device key, which requires an OTP step-up challenge.

Seat reservation

1. The attendee opens a session, then chooses an assigned seat, takes a random seat, or joins open seating with one tap.
2. The API accepts the reservation immediately, with no approval step and no outbound message; the application acknowledges it inline.
3. The hold stays provisional until check-in at the hall gate converts it into a confirmed seat.
4. A sweep shortly before the session releases every hold whose owner has not checked in.

1448/03/20 هـ	تاريخ التطبيق	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج
2027/09/02 م	Date Effective			Form Title
2.0	رقم النسخة			رمز النموذج
إدارة التقنية	مالك النموذج		Apexium	Form Code
	Form Owner			جهة الإصدار
				Issuer

Gate scan

1. A gate operator scans the attendee's QR badge.
2. The API validates the badge, the approval state, and the profile types that gate admits.
3. The API records the scan idempotently and returns an entry or exit outcome.

Live session and moderated questions

1. The attendee opens the session and plays the configured stream.
2. The attendee may submit a question before the session and until it ends.
3. Once live, a hall with a geofence accepts questions only from attendees whose door scan recorded arrival. A hall without one accepts every question.
4. An advisory artificial intelligence (AI) filter tags each question.
5. The Scientific Committee approves, hides or escalates it.
6. The session moderator presents approved questions on stage.

The platform defers automatic arrival detection from device location; the section 3 register records its owner and closing condition.

AI session summary

1. The Control Panel fetches the caption from the published video and hands the subtitle to the API. A team member may paste or upload it instead.
2. The team reviews the subtitle before the API stores it.
3. The API calls the on-site AI service to draft a summary from the subtitle.
4. The Scientific Committee edits and approves the draft, then publishes it only after the session starts.
5. Attendees read the published summary in the application.

Rating and feedback

1. An administrator defines rating types, groups and questions in the Control Panel without a software release.
2. The platform ships five seeded types: App, Session, Forum, Exhibition and Day.
3. Scheduled jobs issue prompts from attendance, per session at its end and per day to that day's arrivals.
4. A closing prompt covers Forum, Exhibition and App.
5. Each type accepts one submission per user; Day accepts one per programme day.

SIMF phase one deployment, security areas and on-site services

UML deployment diagram. Server specifications are reproduced from the customer server requirements workbook.

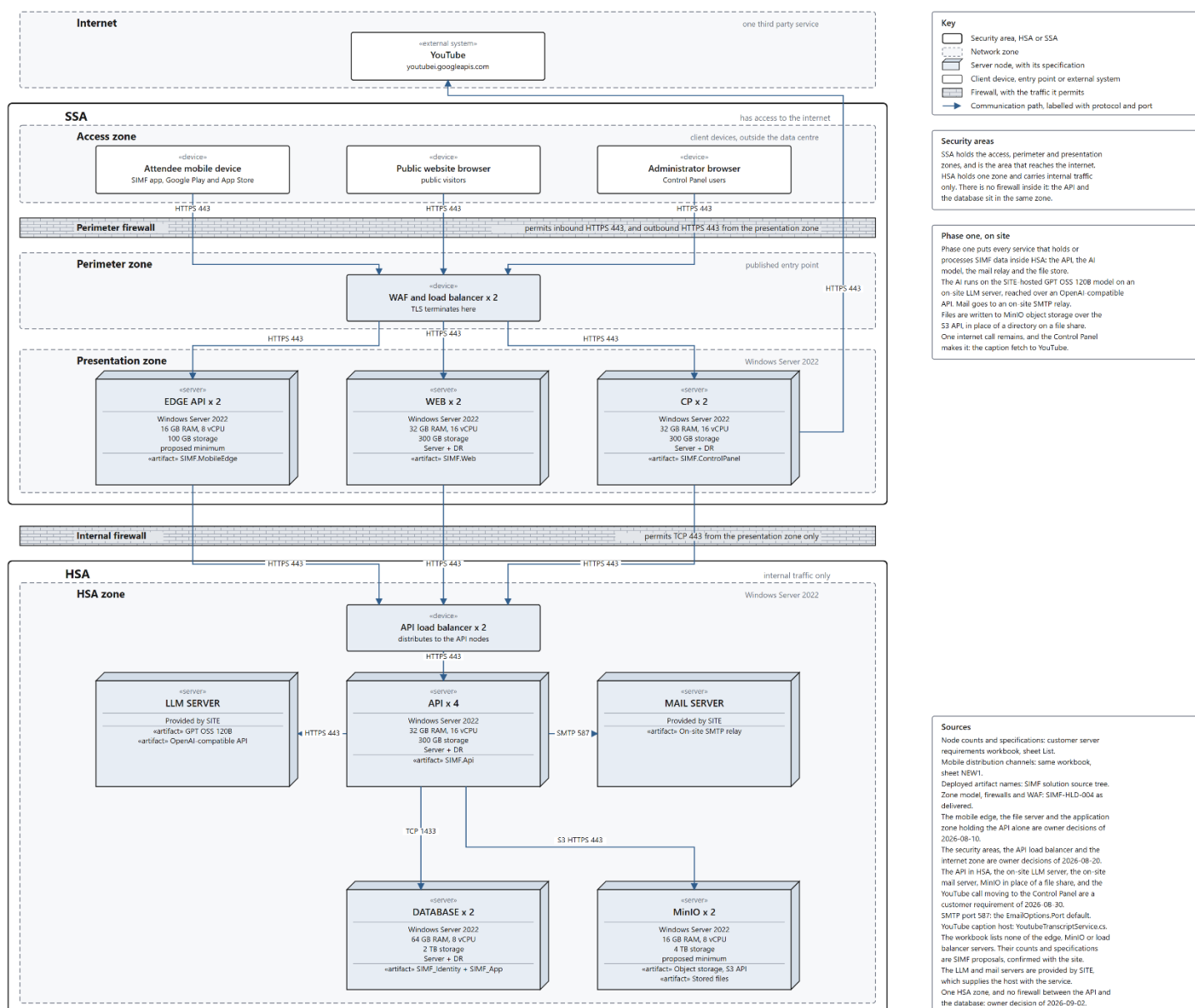


Figure 1. Deployment, security areas and network zones. A UML deployment diagram with one box per hosting server. Each host sits in its network zone, both firewalls carry the rule they enforce, and every connection carries its protocol and port.

SIMF phase one components and their interaction

UML component diagram. Every component carries its technology and every call carries its protocol.

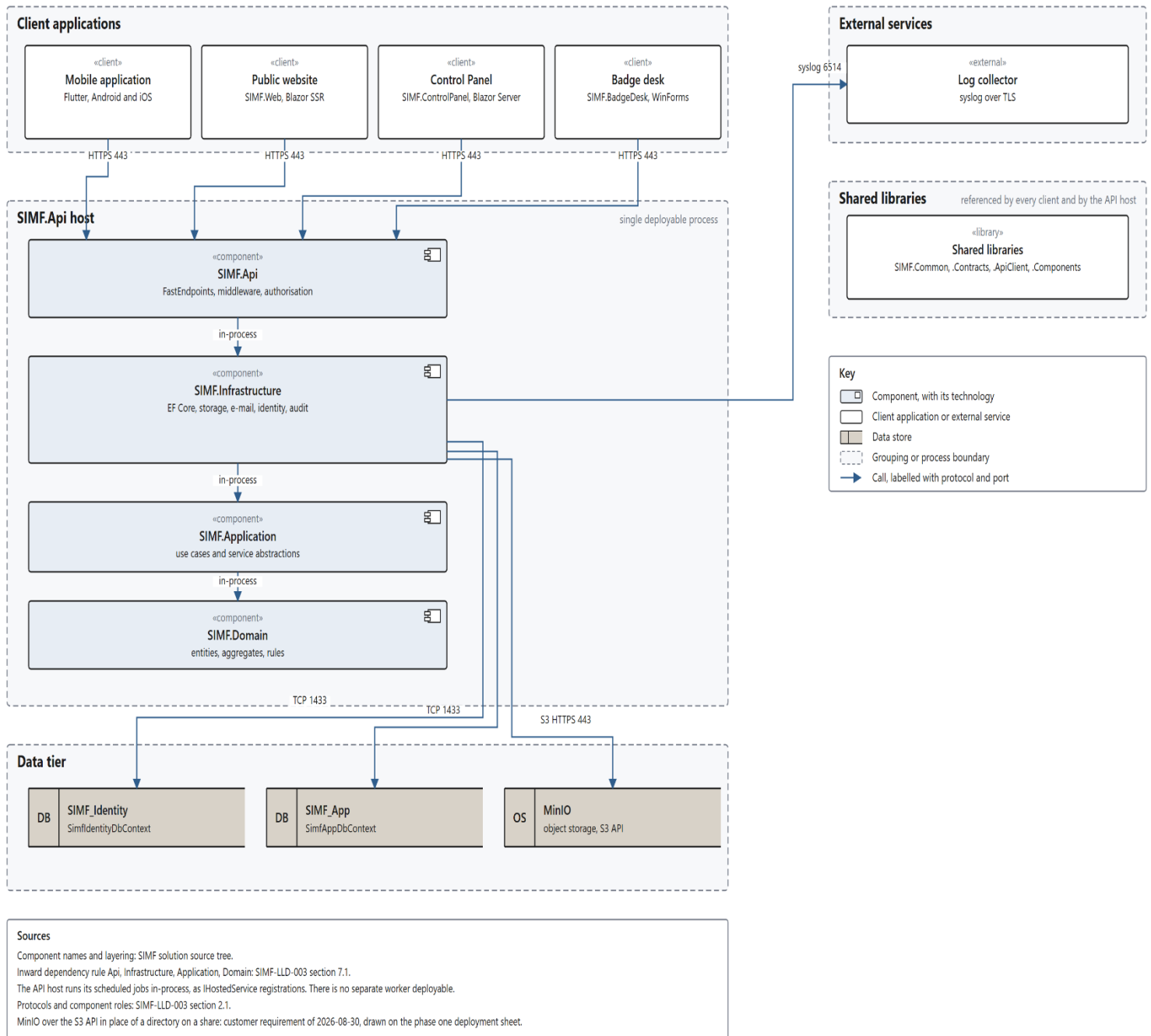


Figure 2. System and data flow. A data flow diagram in Gane and Sarson notation: a level 0 context view of the platform and its external parties, and a level 1 decomposition that balances with it. Every flow that leaves the ministry network is drawn.

SIMF phase one, end-to-end sequence

UML sequence diagram. Ordered messages between the participants of section 2.3, each labelled with its protocol and port.

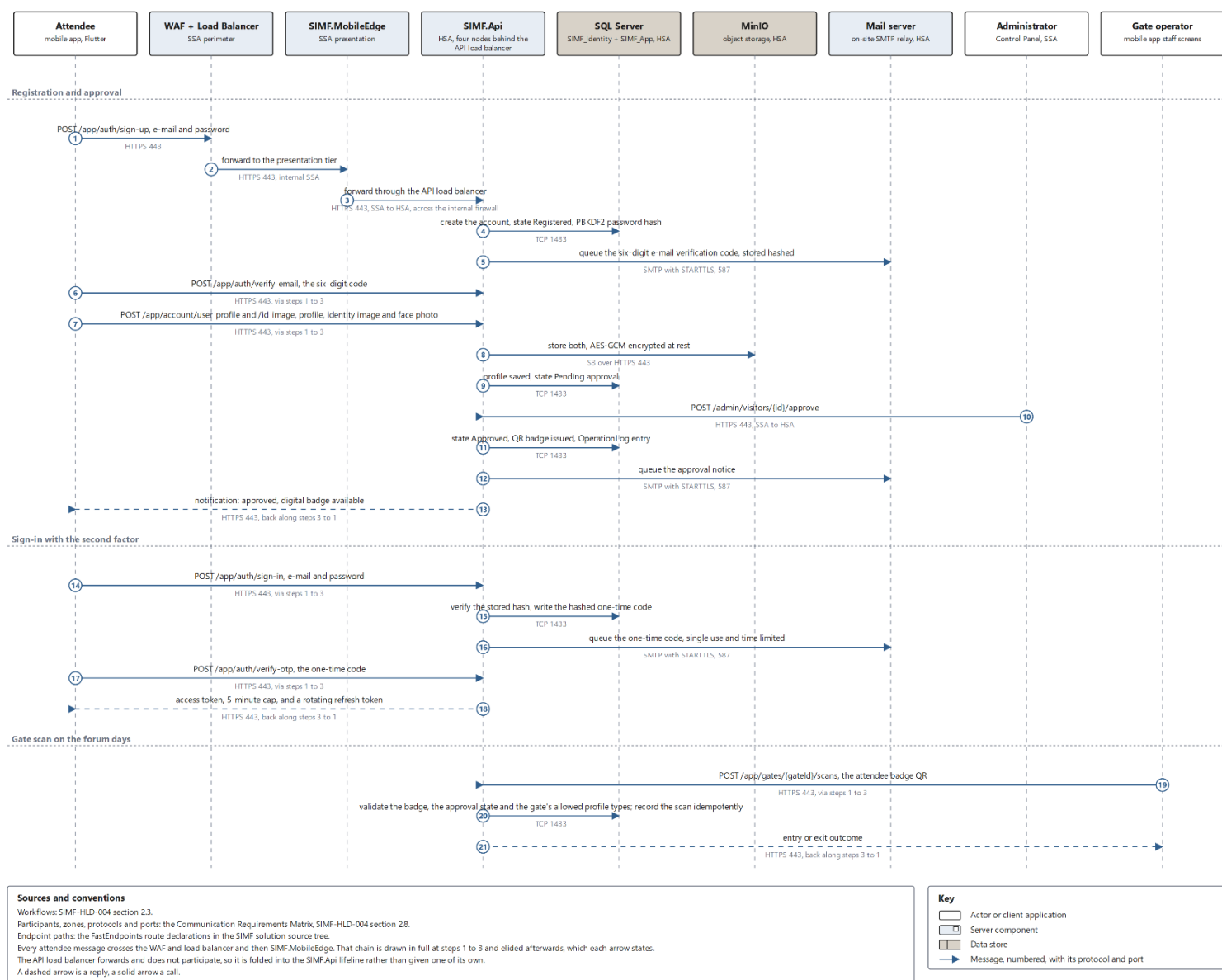


Figure 3. End-to-end sequence. A UML sequence diagram over nine participants, with ordered messages each carrying its protocol and port, covering registration and approval, sign-in with the second factor, and the gate scan.

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م	رقم النسخة Rev. Number			رمز النموذج Form Code
1	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer
إدارة التقنية				

2.4 Integration and Interfaces

The platform sends no personal or confidential data beyond the ministry network.

The platform holds ten interfaces, eight internal, one outbound and one on the attendee's device. The communication matrix in 2.8 carries the protocol and port of each one.

Interface	Direction	Protocol and port	Payload	Authentication	Failure behaviour	Status
Client applications to API	Inbound	(2.8)	ApiResult envelope	JSON Web Token (JWT) bearer, protected routes	Coded bilingual failure envelope	Pending: WAF appliance (SITE)
Control Panel and Website to API	Server to server	(2.8)	Same envelope, typed client	Encrypted cookie, no browser token	Failed envelope, no exception	Agreed
Mobile app to mobile edge	Device to edge	(2.8)	TBD, Apexium	TBD, Apexium	TBD, Apexium	Pending: payload, authentication
Badge desk to admin API	On-site client	(2.8)	One batch per shift	TBD, Apexium	Failure envelope (2.5)	Pending: authentication
API to SQL Server	API to database tier	(2.8)	EF Core, two databases (2.8)	Least-privilege account (2.9)	No silent fallback (2.5)	Pending: quorum witness (SITE)
API to SMTP relay	API outbound	(2.8)	Queued outbound mail	TBD, Apexium	Queued; the request does not wait	Agreed
API to MiniO object store	API outbound, inside HSA	(2.8)	Objects; rows hold keys	TBD, Apexium	No silent fallback (2.5)	Pending: KMS and HSM (SITE)
API to model server	API outbound, inside HSA	(2.8)	Per-feature fields	TBD, SITE	No silent fallback (2.5)	Pending: authentication
YouTube caption import	Outbound to the internet	(2.8)	Public video id only	None, public resource	Administrator supplies the transcript	Pending: egress exception (SITE)
Client-side video playback	Device to YouTube	(2.8)	Stream address only; no media on SIMF servers	None	Playback is independent of the platform	Agreed

Six delivered features call the on-site GPT OSS 120B model, plus a Control Panel prompt test.

Feature	Fields sent	Personal data	Destination
Question filter	Question text only	No	On-site model
FAQ answer	Visitor question, published FAQ entry	No	On-site model
Visitor assistance	Visitor question, public programme content	No	On-site model

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م				
1	رقم النسخة Rev. Number			رمز النموذج Form Code
إدارة التقنية	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer

Subtitle translation	Subtitle text, target language	No	On-site model
Session minutes	Title, speaker display names, abstract, subtitle	Yes, names already public	On-site model
Control Panel assistant	Administrator question, Control Panel content	No	On-site model
Control Panel prompt test	Administrator test prompt	Possible. The prompt is free text typed by an administrator.	On-site model

Every model call carries a bounded field set that the server validates before dispatch. The ceilings sit in the Solution Design Document. The caller's identity never enters a model call. Account identifiers, e-mail addresses, national identity, Iqama and passport numbers, badge identifiers and booking records stay outside the call. Two endpoints from a withdrawn live-translation and sign-language feature remain in the API, and no client calls them. Their removal is an open item in section 3.

The caption call leaves SSA through the perimeter firewall to an allow-list of fully qualified domain names, and every request is logged. The import re-checks the caption host named in the first reply and refuses redirects, blocking server-side request forgery. Without the firewall exception, an administrator pastes or uploads the transcript, and no feature stops. YouTube use covers embedded playback and one caption retrieval; the media team creates and manages broadcasts inside YouTube.

2.5 Error Handling and Logging

Failure model

Central middleware converts domain and validation exceptions into the section 2.4 failure envelope and selects the matching HTTP status code. A validation failure answers 400 and names each offending field in Arabic and English. The API logs an invalid state and raises it; no path silently substitutes a default. In production the API refuses to start when it cannot read a required secret.

Logging and audit

Serilog writes structured events to rolling files, retained locally for 31 days. Every request carries a correlation identifier into the logs and both audit trails. The platform logs seven event classes:

- Authentication, covering sign-in, one-time password (OTP), time-based one-time password (TOTP), token refresh and sign-out
- Authorisation denials
- Rate-limit rejections
- Artificial intelligence invocations
- Security-relevant business events
- Row-level data changes

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م				رمز النموذج Form Code
1	رقم النسخة Rev. Number			
إدارة التقنية	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer

- Unhandled system errors

Two audit trails record who changed what.

Trail	What it records	Attributes captured
OperationLog	Security-relevant business events: approvals, configuration changes, gate scans	Actor, subject, source IP address, user agent, outcome, error code, correlation identifier
RowAudit	Row-level data changes	Operation, before and after images, affected columns, actor snapshot, correlation identifier

Both trails are append-only in the application and at the database: the runtime account may insert rows but may not update or delete them. A host log shipper, configured at deployment, forwards the files and trail entries to the ministry log collector. An actor holding database access therefore cannot erase their own trail.

2.6 User Interface (UI) Design

One design system covers the mobile application, the Control Panel and the public website. All three are fully bilingual: Arabic is the primary language and reads right to left, English the secondary language and reads left to right. The badge desk is a fourth client and sits outside that design system.

Client shells and rendering

Section 2.2 names each client's technology; this section states only the shell and rendering model.

- Mobile application: a persistent tab shell carries the navigation.
- Control Panel: interactively rendered; every list page uses one standard data grid above a shared create, read, update and delete framework.
- Public website: statically rendered, with interactive elements confined to a small number of pages.
- Badge desk: a Windows Forms desktop client, covered in section 2.2.

Tokens, layout and accessibility

Colour, typography, spacing and corner radius are defined once and shared by all three clients. Every layout starts Arabic first in right-to-left form and mirrors automatically for English. Widths are responsive rather than fixed, for phone and tablet use. The website targets Web Content Accessibility Guidelines (WCAG) level AA. The mobile client lets the user set text size, high contrast and reduced motion. The screen and route inventory for every client sits in section 4 of the Solution Design Document. The minimum supported browser and mobile operating system versions remain open; section 3 carries them in the open items register.

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م				رمز النموذج Form Code
1	رقم النسخة Rev. Number			
إدارة التقنية	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer

2.7 Operational Model View

Environments

SIMF runs on four environments, and every change reaches production by passing through each of them in order.

The four environments differ in topology, database edition, node allocation and purpose.

Environment	Topology	Database edition	Nodes	Purpose
Development	One node hosting all four sites	SQL Server Standard, one instance	One application node, one database instance	Feature development, with the OpenAPI description exposed
Test	One node hosting all four sites	SQL Server Standard, one instance	One application node, one database instance	Integrated verification before promotion, with the OpenAPI description exposed
Staging	The production topology, reproduced	As production	As production; counts TBD, owner SITE operations, closing when SITE issues them	Pre-production rehearsal of the live service
Production	Three tiers, presentation, application and data, across two security areas, shown in Figure 1	SQL Server 2022 Enterprise in an AlwaysOn Availability Group	As section 2.1 specifies	Live service

Staging fills the pre-production role. It must be available two weeks before launch, and the load test runs there. Promotion skips no environment. A change is promoted only after a clean Release build carrying zero warnings. A build warning, or a dependency with a known vulnerability, fails the pipeline and halts promotion.

Zones and network addressing

Figure 1 and the section 2.1 zone table allocate every production component to its security area, SSA or HSA. The same table allocates each component to its network zone. It also carries the per-tier hardware specification. Network addressing and virtual local area network (VLAN) allocation are not yet assigned; section 3 carries them as OI-09.

Production prerequisites

Four items must be in place before go-live, and each carries a named owner. A fifth item is recommended rather than required.

1. A certificate authority signed TLS certificate covering the published SIMF host names. Owner: SITE operations.
2. MinIO object storage provisioned inside HSA. Owner: SITE operations.

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م				رمز النموذج Form Code
1	رقم النسخة Rev. Number			
إدارة التقنية	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer

3. One shared data-protection key ring folder that the website nodes and the Control Panel nodes can read and write. Owner: SITE operations. Neither site starts without it.

4. Session affinity enabled on the website and Control Panel tiers. Owner: SITE operations.

5. Recommended: an outbound exception permitting the Control Panel hosts, and no other host, to reach the YouTube caption hosts. Owner: the site network team. Section 2.4 states what the platform does when the exception is refused. Section 3 carries it as OI-07.

Service operations

Production hosts four Internet Information Services (IIS) sites, each with its own application pool. The four sites are the API, the Control Panel, the website and the mobile edge. Each site deploys independently of the other three, so one site restarts without stopping the others. Scheduled jobs run inside the API process, as section 2.2 describes, so restarting them means recycling the API application pool. Each node reads its signing key and its two data keys from the key management service (KMS) at start-up. Section 2.5 gives the behaviour when a key cannot be read, and it states how log files reach the ministry log collector.

2.8 Data Architecture View

Logical data groups

Two physically separate SQL Server 2022 databases hold all persistent data. The split is permanent and isolates the identity surface. SIMF_Identity holds accounts, roles and permissions, token and second-factor material, device keys and password history. SIMF_App holds business data: profiles, the programme, seating and gate activity, published content, networking, feedback, configuration and both audit trails.

Separation invariants

These three rules are permanent.

- No foreign key spans the two databases; a business record names a user by a bare identifier, resolved by a second query.
- Live data is never duplicated; the one permitted copy is an immutable audit snapshot of the actor's name and e-mail.
- No transaction spans both databases, and each connection string may address one instance or two servers.

Data flow

TLS terminates at the WAF and load balancer, which routes each client to its presentation component. Those components call SIMF.Api through the API load balancer, crossing the internal firewall. The API authorises the caller, then reaches the databases, file store, mail relay and inference endpoint. The badge desk workstation sits on the venue network, outside the SSA and the HSA. It uploads each shift through the WAF and load balancer; section 2.1 states the firewall rule that governs this path. Figure 2 shows these flows and their trust boundaries.

File store

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م	رقم النسخة Rev. Number			رمز النموذج Form Code
1	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer
إدارة التقنية				

The file store holds binary content kept out of the database and readable by any node. MinIO serves it over the S3 API; database rows carry only object keys. Every read and write passes through the API, which applies the content's category policy; no client reaches the store. Four sensitivity tiers classify stored content. Section 2.9 states which content is encrypted at rest. The tier definitions and the rule that assigns content to a tier are TBD, owner Apexium. The closing condition is the file classification published in the Solution Design Document. Session recordings rely on access control, because byte-range playback needs seekable content.

Integrity and validation

- Primary keys, foreign keys and unique indexes inside each database prevent duplication.
- Each unit of work is atomic within one database, so no partial update survives.
- The software engineering standards document governs server-side input validation and parameterised data access.
- Section 2.9 states the encryption posture, at rest and in transit.
- Business records are deactivated rather than deleted, preserving referential history; section 2.5 defines the audit trails.

Concurrent sessions and throughput are stated in section 2.1.

Communication requirements matrix

The matrix below records every flow drawn on Figure 1.

Source	Destination	Protocol	Port	Direction	Purpose
Mobile app, Internet	WAF and load balancer	HTTPS, TLS 1.2+	443	Inbound	Application traffic
Public browser, Internet	WAF and load balancer	HTTPS, TLS 1.2+	443	Inbound	Website traffic
Administrator, Internet	WAF and load balancer	HTTPS, TLS 1.2+	443	Inbound	Control Panel traffic
Badge desk workstation, venue network	WAF and load balancer	HTTPS, TLS 1.2+	443	Inbound	Badge desk shift batch upload
WAF and load balancer	SIMF.MobileEdge	HTTPS	443	Internal, SSA	Mobile edge requests
WAF and load balancer	SIMF.Web	HTTPS	443	Internal, SSA	Website rendering
WAF and load balancer	SIMF.ControlPanel	HTTPS	443	Internal, SSA	Administration
SIMF.MobileEdge, SIMF.Web, SIMF.ControlPanel	API load balancer	HTTPS	443	SSA to HSA	Presentation to API
API load balancer	SIMF.Api	HTTPS	443	Internal, HSA	API node distribution
SIMF.Api	GPT OSS 120B server	HTTPS	443	Internal, HSA	Model inference
SIMF.Api	Mail server	SMTP with STARTTLS	587	Internal, HSA	Transactional e-mail
SIMF.Api	SQL Server Availability Group listener	TCP, TDS	1433	Internal, HSA	Both databases
SIMF.Api	MinIO object store	HTTPS, S3 API	443	Internal, HSA	File access

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م	رقم النسخة Rev. Number			رمز النموذج Form Code
1	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer
إدارة التقنية				

SQL Server replica	SQL Server replica	TCP	5022	Internal, HSA	Availability Group replication
SIMF.ControlPanel	YouTube caption hosts	HTTPS	443	Outbound	Caption import, video identifier only
Application hosts	Ministry log collector	Syslog over TLS	6514	Outbound	Centralised structured logging
Attendee device, Internet	YouTube	HTTPS	443	Client-side	Live playback, no SIMF server

2.9 Solution and Non-Functional Views

Security

Every audience signs in with an e-mail address and a password. The platform stores a password only as a one-way hash, and can force a change before sign-in completes. Visitors confirm with a one-time password (OTP) sent by e-mail, administrators with a time-based one-time password (TOTP) or a single-use recovery code. No token is issued until the second factor succeeds, and the platform never e-mails a password. Mobile sign-in may add a device key, generated inside the device hardware security element and never exported. Only its public half is registered against the account and device. An administrator can revoke a device, after which the server refuses its key.

Access tokens are JSON Web Tokens (JWT), signed asymmetrically rather than with a shared secret. Only the API inside HSA holds the private key, so no presentation host can mint a token. Access tokens are short-lived, refresh tokens rotate on use, and reuse is detected. An absolute session cap applies, and a security stamp revokes a session at once. The Solution Design Document carries the token lifetimes and the signing algorithm. Authorisation is role-based access control (RBAC), with per-page and per-action permissions from one catalogue, enforced on the API and the Control Panel alike.

A web application firewall (WAF) at the edge blocks injection, cross-site scripting and cross-site request forgery. All traffic runs on TLS 1.2 or higher, terminated at the load balancer and re-encrypted onward. Section 2.4 states the authentication each internal interface carries. AES-256-GCM protects the encrypted file categories and the national identity, Iqama, passport and mobile number columns. Those keys and the application secrets live in the key management service (KMS) behind a hardware security module (HSM), never in configuration.

Rate limiting applies on several dimensions, not one global rule, and every rejection reaches the audit trail. The application sets security headers and a content security policy, and allows cross-origin resource sharing (CORS) only from an explicit origin list. It scans every uploaded file. Text sent to the on-site model is sanitised against hidden instructions. Database accounts hold least privilege, and the runtime account holds data rights only.

Section 2.5 describes the two audit trails.

An offline server-side check inside HSA confirms an uploaded photograph holds a human face, so no badge is issued against an unusable picture. The mobile application runs a guided liveness capture wholly on the device, producing the photograph and nothing else. Optional biometric sign-in asks the operating system to check a fingerprint or face,

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م	رقم النسخة Rev. Number			رمز النموذج Form Code
1	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer
إدارة التقنية				

which unlocks the device key and never reaches SIMF. The platform performs no facial recognition, computes no face embedding, stores no biometric template and matches no face against any register.

The platform meets its Saudi Personal Data Protection Law (PDPL) obligations as follows.

Obligation	How the platform meets it	Control point
Lawful basis	The registrant enters their own data and accepts the terms	Registration form
Purpose limitation	Personal data serves event admission only, and is never profiled, sold or shared	Access control
Minimisation	Fields are limited to what a badge and a security approval require	Registration form
Access and correction	The holder reads and edits their profile, and may delete the photograph alone	Profile screen
Erasure	Irreversible erasure hard deletes the personal data and files, revokes sessions and device keys, and disables the account; the business record survives without identifying data, an exception to the deactivation rule in section 2.8	Mobile application, any account state
Residency	Every store holding personal data sits inside the HSA zone	Hosting zones, Figure 1
Retention and disposal	Personal data is scoped to the edition it was collected for and disposed of through the erasure path; the disposal schedule is open, item OI-05	Ministry data-protection function

The design conforms to its mandated standards as follows.

Standard	Conformance	Open items
NCA Secure Application Development Standard	Designed to conform	Independent penetration test not yet commissioned, item OI-01, section 3
NCA Essential Cybersecurity Controls (ECC)	Conforms except on identity lifecycle	Password expiry ships inactive; enabling it, secret rotation and the signed certificate are pre-launch activities, item OI-01, section 3
Saudi Personal Data Protection Law	Designed to conform	Retention schedule not yet agreed, item OI-05, section 3
OWASP application security baseline	Designed to conform	WAF appliance and ruleset agreed at deployment, section 2.4

1448/02/02 هـ	تاريخ التطبيق	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج
2027/07/21 م	Date Effective			Form Title
1	رقم النسخة			رمز النموذج
	Rev. Number			Form Code
إدارة التقنية	مالك النموذج		Apexium	جهة الإصدار
	Form Owner			Issuer

Availability

Database availability rests on a SQL Server 2022 Enterprise AlwaysOn Availability Group (AG) holding both databases. It runs one read-write primary and one readable secondary, replicates by synchronous commit, and fails over automatically. The listener routes write to the primary and read-only queries to the secondary. A cluster quorum witness completes the cluster; its configuration is open in section 3.

A valid restore needs four artefacts from one point in time. They are the two databases, the object store contents, and an escrowed copy of the encryption keys. The keys belong in that set because encrypted columns and files restore unreadable without them. Backups run on a schedule, and a set is restored into a non-production environment on a defined cycle. Cadence, retention and the rehearsal interval are TBD, owner SITE. The recovery site and its replication method remain open in section 3.

The platform commits to the following availability, recovery point and recovery time objectives.

Objective	Commitment	Basis
Availability during the three forum days	99.9 percent	An outage inside the event window cannot be rescheduled. Time inside a node or database recovery, bounded by the recovery time objective below, is excluded; the permitted number of such events per window is TBD, owner SITE, confirmed in the deployment plan
Availability outside the event window	99.5 percent	Normal service
Recovery point objective, databases, node or replica failure	Zero	The Availability Group runs synchronous commit, so a committed transaction is on both replicas before the commit returns
Recovery point objective, object store	15 minutes	Snapshot interval, confirmed with SITE during deployment planning
Recovery point objective, loss of the HSA zone	The backup cadence, which cannot better the 15-minute object store interval	A zone loss restores from the four artefact backup set, so the exposure is the gap between backups
Recovery time objective, node or database failure	5 minutes	Availability Group failover is automatic and takes seconds; the budget covers detection, load balancer rotation and client retry
Recovery time objective, loss of the HSA zone	4 hours	A restore from the four artefact backup set, not a failover

Monitoring

Each host writes structured events to rolling files, each carrying a correlation identifier for tracing. Section 2.7 states how they reach the ministry log collector, and administrators read both audit trails in the Control Panel. A health endpoint serves as the readiness probe and reports the scheduled jobs from their heartbeat registry. A faulted job turns the

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م				
1	رقم النسخة Rev. Number			رمز النموذج Form Code
إدارة التقنية	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer

node degraded and drops it from rotation. A monitoring toolchain collects processor, memory, response-time, throughput and query metrics, and alerts on thresholds. The Control Panel presents usage statistics for registrations, attendance, bookings, scans and ratings. The alerting product, its thresholds and the escalation path are TBD, owner SITE.

Sizing and Performance

Scaling out is a deployment change, not a redesign: nothing scales itself, and capacity is added node by node. The website scales horizontally behind the load balancer. The API is stateless, so nodes are added freely for registration surges, gate-scan bursts and live-session load. Its baseline is four nodes, sized in section 2.1; the database tier scales vertically instead.

Queries are indexed and projected, and list endpoints page at the data source rather than in memory. Live notifications and questions use bounded conditional polling, and this build carries no server-push transport. At the peak concurrency given in section 2.1, polling is roughly 100 requests per second across the four API nodes. The Solution Design Document carries the poll interval, the cache behaviour and the response codes. Reference data is cached, and large media is served through a caching layer rather than the API. Load tests in staging set the pass thresholds, which section 3 carries.

3. Risks and Mitigation Strategies

Six risks govern delivery. Likelihood is the residual assessment once the mitigation is in place, and each mitigation names the section that implements it.

The table below sets out each risk with its impact, likelihood, mitigation and owner.

Risk	Impact	Likelihood	Mitigation	Owner
Security	Exposure of attendee identity documents, or an unauthorised administrative action	Low	Permission gating per page and action, encryption at rest and in transit, a web application firewall, rate limiting, key management service secrets, and pre-handover penetration testing (2.9)	Apexium
Event-day availability	Loss of registration, gate entry and live-session access during the forum window, against the constraints in 1.2	Low	Load balancer, stateless multi-node application tier, Availability Group failover, health-probe rotation, and rollback to the last good build (2.7, 2.9)	SITE
Scalability and performance	Registration surge, live-session concurrency, gate-scan bursts and notification fan-out coincide and slow responses	Medium	Scale-out per tier, indexed and paged queries, bounded polling with caching, and peak-shaped staging load tests (2.9)	Apexium
Supplier and technology dependency	A SITE-supplied service is delayed or withdrawn: hosting, mail relay, model server or recovery site	Medium	Inference runs on site, so no external provider sits in the path; the provider is one configurable abstraction (2.4)	SITE

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م	رقم النسخة Rev. Number			رمز النموذج Form Code
1	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer
إدارة التقنية				

Data protection	Loss or corruption of attendee, booking or scan data, or no evidence of who changed it	Low	Separated databases with no cross-database key or transaction, database-enforced integrity, soft delete, reconciliation of logical references, and append-only audit trails (2.5, 2.8)	Apexium
Schedule and change freeze	A defect found after the freeze is expensive and disruptive to correct	Medium	Four-stage promotion with no stage skipped, reviewed pull requests, a pipeline that halts on a warning or vulnerable dependency, and per-page regression scenarios (2.7)	Apexium

Open items register

The register below is the only list of open items in the design.

Identifier	Item	Owner	Closing condition
OI-01	Pre-launch deployment activities listed in 2.7 and 2.9	SITE	Each activity completed and verified in production before go-live
OI-02	Venue boundary for the deferred geofence behaviour	RSNF	Boundary confirmed, then the behaviour designed and scheduled
OI-03	The statistics metric list	RSNF	Metric list confirmed in writing
OI-04	Load test pass thresholds	Apexium	Thresholds agreed, then met by a peak-shaped staging run
OI-05	Personal data retention schedule	RSNF	Retention period approved for each category of personal data
OI-06	Availability Group quorum witness	SITE	Witness type and location confirmed and configured
OI-07	Outbound exception for the caption import	SITE	Rule approved and in place, or transcript upload adopted instead
OI-08	Proposed sizing for the mobile edge and object store hosts	SITE	Node count and specification agreed with the customer
OI-09	Network addressing and virtual local area network allocation	SITE network team	Addresses and identifiers issued for every zone
OI-10	Recovery site, and the replication method for the databases and the object store	SITE	Site named and replication confirmed against the four-hour recovery time objective
OI-11	Minimum supported browser and mobile operating system versions	Apexium	Confirmed from the delivered client build configuration

Pre-launch activities close before go-live; the rest stay open until their closing condition is met.

1448/02/02 هـ	تاريخ التطبيق Date Effective	 وزارة الدفاع MINISTRY OF DEFENSE	HLD Form	عنوان النموذج Form Title
2027/07/21 م				رمز النموذج Form Code
1	رقم النسخة Rev. Number			
إدارة التقنية	مالك النموذج Form Owner		Apexium	جهة الإصدار Issuer

Approvals

Technology Expert, Role 1

Signature: _____

Date_____

Technology Head

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Date_____

Technology Director

Signature: _____

Date_____